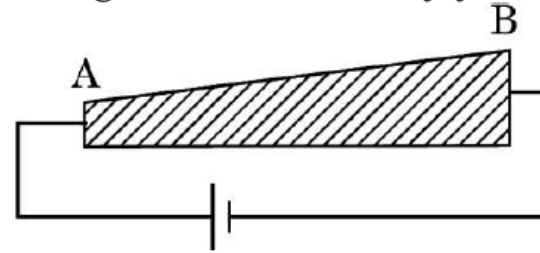


- 26.** Name the factors on which electrical conductivity of a material depends. Obtain the relation between current density in a conductor and the conductivity of its material.

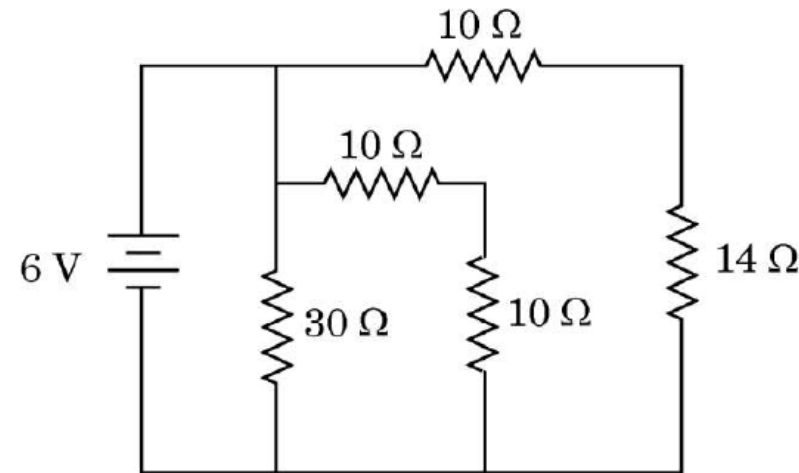
19. The terminal potential difference of a cell is 19 V when a current of 1.0 A flows in the circuit. It reduces to 17 V when the current supplied by the cell is 3.0 A. Find the emf and internal resistance of the cell.

2. A current flows through a series combination of two copper wires of equal length but their radii are in the ratio of 1 : 2. The ratio of drift velocities of free electrons in the wires will be :
- (a) 8 (b) 4 (c) 2 (d) 1

32. (a) (i) Define mobility of electrons. Give its SI units.
- (ii) A steady current flows through a wire AB, as shown in the figure. What happens to the electric field and the drift velocity along the wire? Justify your answer.



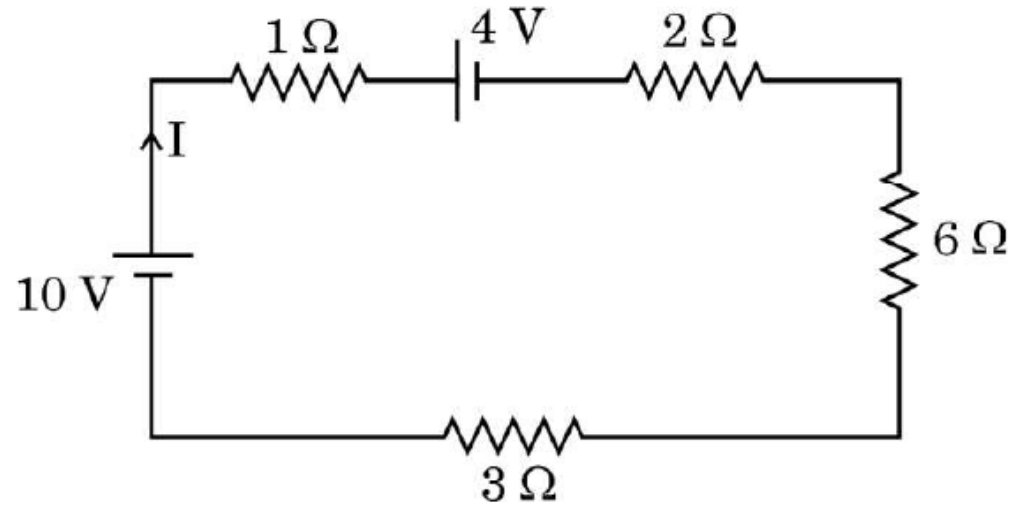
- (iii) Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery.



OR

- (b) (i) Define electrical conductivity of a wire. Give its SI unit.
- (ii) High current is to be drawn safely from (1) a low-voltage battery, and (2) a high-voltage battery. What can you say about the internal resistance of the two batteries ?
- (iii) Calculate the total energy supplied by the batteries to the circuit shown in the figure, in one minute.

5



22. A cell of emf E and internal resistance r is connected to a variable resistance R . Draw plots showing the variation of (a) terminal voltage V with R , and (b) V with current I , in the circuit.

17. *Assertion (A) :* When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source, the power consumed by the 50 W bulb is maximum.

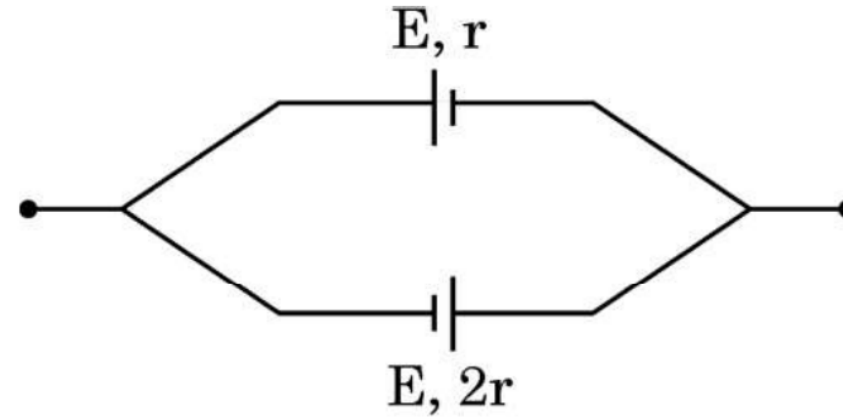
Reason (R) : In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.

- 25.** Two identical cells, each of emf E and internal resistance r , are connected with a load resistance R , first in series and then in parallel. Obtain the condition under which the current through R is same in both cases. 2

20. Write two differences between the emf and terminal potential difference of a cell. What is the most important precaution that one should take while drawing current from a cell ?

1. A battery of emf E and internal resistance r is connected to an external circuit. The potential drop within the battery is proportional to :
- (a) current in the circuit
 - (b) total resistance of the circuit
 - (c) emf of the battery
 - (d) power dissipated in the circuit

14. Two cells of emf E each and internal resistances r and $2r$ are connected in parallel as shown in the figure. The equivalent emf of the combination is :



(a) zero

(b) $\frac{E}{2}$

(c) $\frac{E}{3}$

(d) E

17. *Assertion (A) :* The internal resistance of a cell is constant.

Reason (R) : Ionic concentration of the electrolyte remains same during use of a cell.

8. The SI unit of mobility of charge carriers is :

(a) $\Omega \text{ s}^{-1}$

(b) $\text{m}^2 \text{ V}^{-1} \text{ s}^{-1}$

(c) $\text{m s}^{-1} \text{ V}^{-1}$

(d) $\Omega \text{ m}$

13. The emf and internal resistance of a cell are E and r respectively. It is connected across an external resistance $R = 2r$. The potential drop across the terminals of the cell will be :

(a) $\frac{E}{4}$

(b) $\frac{E}{2}$

(c) $\frac{2}{3}E$

(d) $\frac{E}{3}$

30. A potential difference ' V ' is applied across a load resistor of resistance R . V and R can be varied. If the current that flows in the circuit is I , draw a plot showing the variation of power consumed by the resistor as a function of :

3

- (a) R , keeping V constant
- (b) I , keeping R constant
- (c) V , keeping R constant

3. The current density due to drift of electrons in a conductor is given by (symbols have their usual meanings)

(a) $n e A v_d$

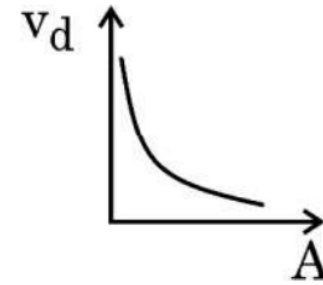
(b) $\frac{n A v_d}{e}$

(c) $\frac{n v_d}{e A}$

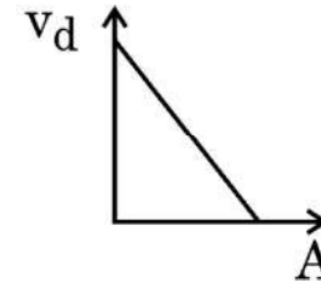
(d) $n e v_d$

26. A potential difference V is applied across a conductor of length l and uniform cross-section area A . How will the (i) electric field E , (ii) drift velocity v_d , and (iii) current density j be affected when (a) V is doubled and (b) l is halved (keeping other factors constant) ?

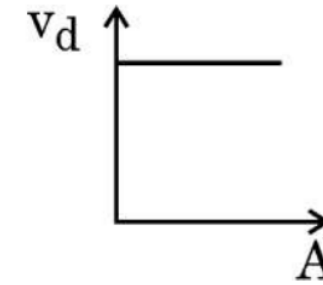
2. A steady current flows through a metallic wire whose area of cross-section (A) increases continuously from one end of the wire to the other. The magnitude of drift velocity (v_d) of the free electrons as a function of ' A ' can be shown by :



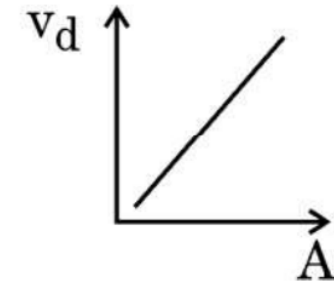
(a)



(b)

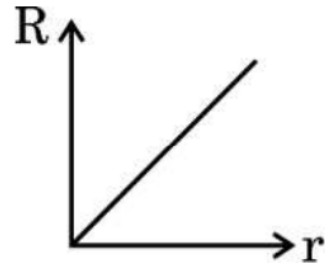


(c)

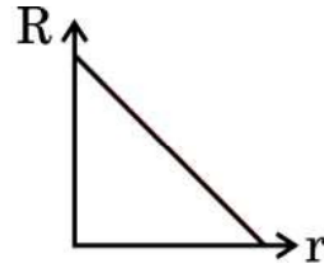


(d)

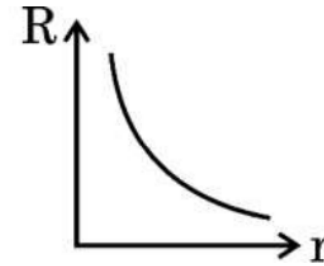
15. The correct graph showing the variation of the resistance (R) of a cylindrical metal wire as a function of its radius (r), keeping its length and temperature constant, is :



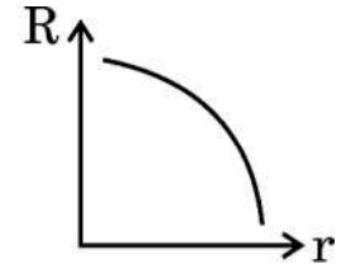
(a)



(b)

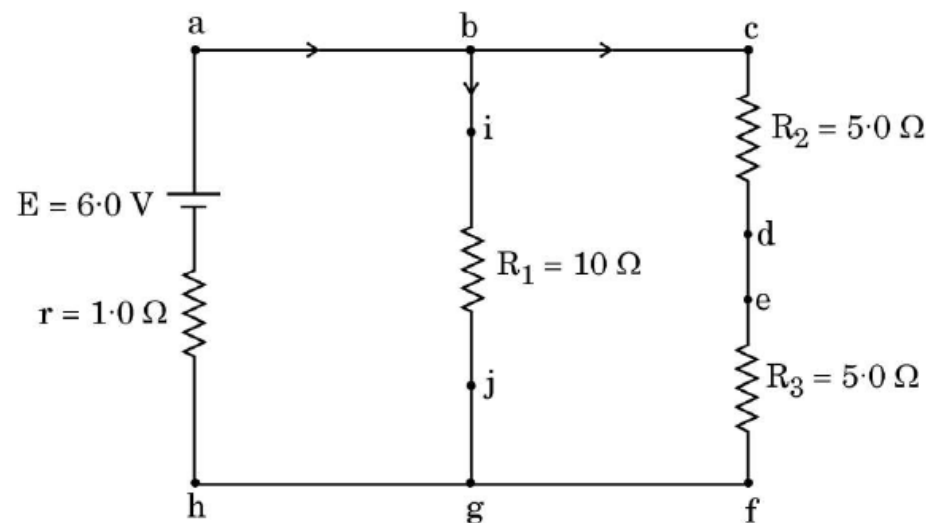


(c)



(d)

34. The following figure shows a circuit diagram. We can find the currents through and potential differences across different resistors using Kirchhoff's rules.



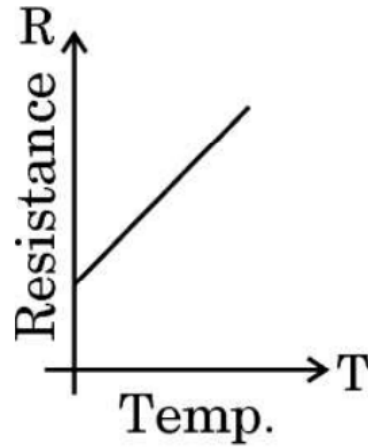
Answer the following questions based on the above :

- | | |
|---|---|
| (a) Which points are at the same potential in the circuit ? | 1 |
| (b) What is the current through arm bg ? | 1 |
| (c) Find the potential difference across resistance R_3 . | 2 |

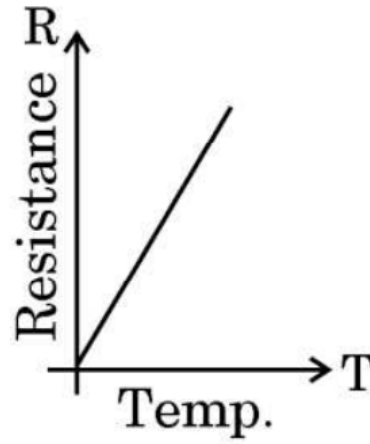
OR

- | | |
|--|---|
| (c) What is the power dissipated in resistance R_2 ? | 2 |
|--|---|

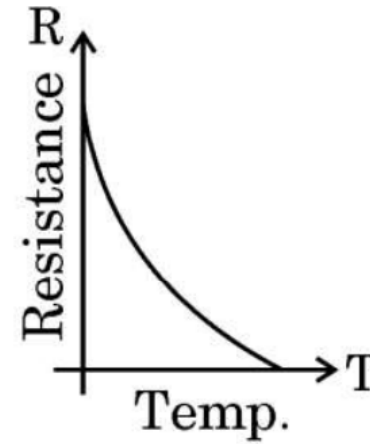
2. For a metallic conductor, the correct representation of variation of resistance R with temperature T is :



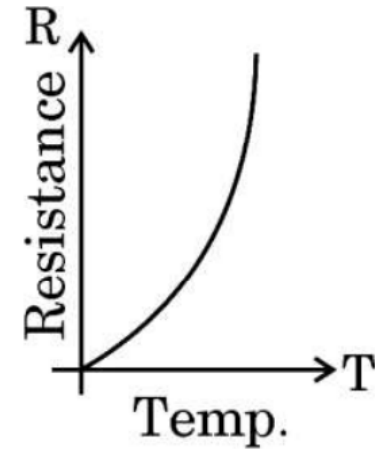
(a)



(b)



(c)



(d)

11. The current in a device varies with time t as $I = 6t$, where I is in mA and t is in s. The amount of charge that passes through the device during $t = 0$ s to $t = 3$ s is

1

(a) 10 mC

(b) 18 mC

(c) 27 mC

(d) 54 mC

13. A cell of emf E is connected across an external resistance R . When current 'I' is drawn from the cell, the potential difference across the electrodes of the cell drops to V . The internal resistance 'r' of the cell is

1

(a) $\left(\frac{E - V}{E}\right) R$

(b) $\left(\frac{E - V}{R}\right)$

(c) $\frac{(E - V) R}{I}$

(d) $\left(\frac{E - V}{V}\right) R$

29. A potential difference V is applied across a conductor of length l and cross-sectional area A . Briefly explain how the current density j in the conductor will be affected if
- (a) the potential difference V is doubled,
 - (b) the conductor were gradually stretched to reduce its cross-sectional area to $\frac{A}{2}$ and then the same potential difference V is applied across it.

7. A steady current of 8 mA flows through a wire. The number of electrons passing through a cross-section of the wire in 10 s is 1
- (a) 4.0×10^{16} (b) 5.0×10^{17}
- (c) 1.6×10^{16} (d) 1.0×10^{17}

9. A conductor of $10\ \Omega$ is connected across a $6\ \text{V}$ ideal source. The power supplied by the source to the conductor is 1
- (a) $1.8\ \text{W}$ (b) $2.4\ \text{W}$
(c) $3.6\ \text{W}$ (d) $7.2\ \text{W}$

29. Two cells of emf E_1 and E_2 and internal resistances r_1 and r_2 are connected in parallel, with their terminals of the same polarity connected together. Obtain an expression for the equivalent emf of the combination. **3**

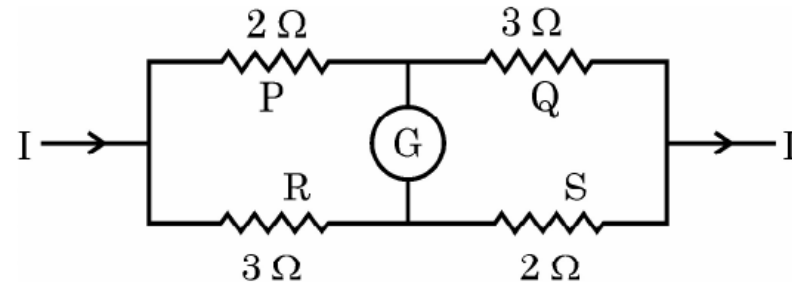
3. A current of 0.8 A flows in a conductor of $40\ \Omega$ for 1 minute. The heat produced in the conductor will be 1
- (a) 1445 J (b) 1536 J
- (c) 1569 J (d) 1640 J

14. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T . The conductivity of

1

- (a) both increases.
- (b) both decreases.
- (c) copper increases and silicon decreases.
- (d) copper decreases and silicon increases.

17. **Assertion (A)** : The given figure does not show a balanced Wheatstone bridge.



Reason (R) : For a balanced bridge small current should flow through the galvanometer.

27. Define current density and relaxation time. Derive an expression for resistivity of a conductor in terms of number density of charge carriers in the conductor and relaxation time.

24. Two conductors, made of the same material have equal lengths but different cross-sectional areas A_1 and A_2 ($A_1 > A_2$). They are connected in parallel across a cell. Show that the drift velocities of electrons in two conductors are equal.

33. (a) (i) Explain how free electrons in a metal at constant temperature attain an average velocity under the action of an electric field. Hence obtain an expression for it. **3**
- (ii) Consider two conducting wires A and B of the same diameter but made of different materials joined in series across a battery. The number density of electrons in A is 1.5 times that in B. Find the ratio of drift velocity of electrons in wire A to that in wire B. **2**

OR

- (b) (i) A cell emf of (E) and internal resistance (r) is connected across a variable load resistance (R). Draw plots showing the variation of terminal voltage V with (i) R and (ii) the current (I) in the load.

2

- (ii) Three cells, each of emf E but internal resistances $2r$, $3r$ and $6r$ are connected in parallel across a resistor R .

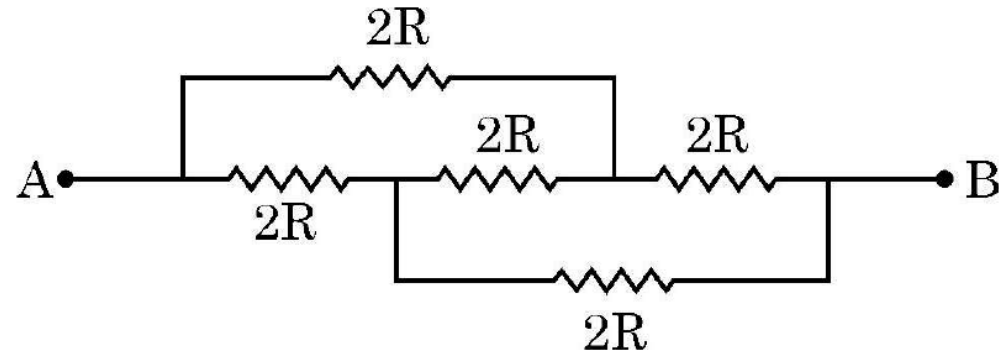
Obtain expressions for (i) current flowing in the circuit, and (ii) the terminal potential difference across the equivalent cell.

3

17. **Assertion (A)** : The equivalent resistance between points A and B in the given network is $2R$.

1

Reason (R) : All the resistors are connected in parallel



24. A potential difference (V) is applied across a conductor of length ' L ' and cross-sectional area ' A '.

How will the drift velocity of electrons and the current density be affected if another identical conductor of the same material were connected in series with the first conductor ? Justify your answers.

24. The potential difference applied across a given conductor is doubled. How will this affect (i) the mobility of electrons and (ii) the current density in the conductor ? Justify your answers.

3. The amount of charge flowing across a cross-section of a conductor, connected to a battery, in 4.0 s is 720 mC. The current in the circuit is :

(a) 0.18 A

(b) 1.8 A

(c) 18 mA

(d) 18 A

4. Two cells, one of emf $2E$ and internal resistance r and the other of emf E and internal resistance $r/2$ are connected in parallel, by connecting their positive terminals together and their negative terminals together. The equivalent emf of the combination is :

(a) $3E$

(b) $\frac{4E}{3}$

(c) $2E$

(d) $\frac{3E}{4}$

- 26.** A cylindrical conductor of resistance R is connected to a battery. The drift velocity of electrons in this conductor is v_d . The conductor is disconnected from the battery and gradually stretched so that its length increases by 25%. It is again connected to the same battery. Find the new value of : 3
- (a) the drift speed of electrons, and
 - (b) the resistance of the conductor.

16. *Assertion (A) :* The temperature coefficient of resistance is positive for metals and negative for semi-conductors.

Reason (R) : The charge carriers in metals are negatively charged whereas in semiconductors they are positively charged.

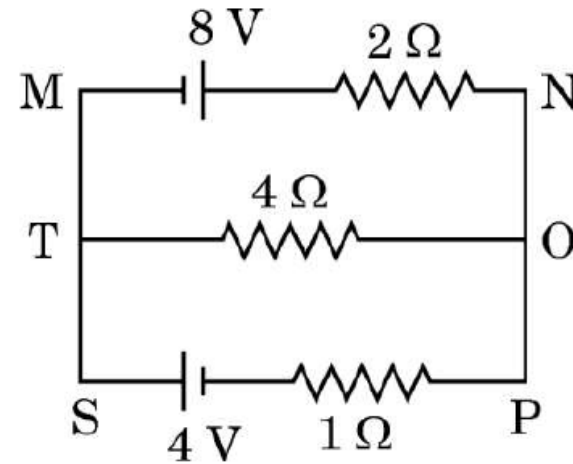
25. (a) When is more power delivered to a light bulb — just after it is turned on and the glow of the filament is increasing or after the glow becomes steady ? Why ? 2

OR

- (b) A battery is connected first across the series combination and then across the parallel combination, of three resistances R , $2R$ and $3R$. In which of the three resistances will power dissipated be maximum in the two cases ? Justify your answer. 2

- (b) (i) State Kirchhoff's rules. Use them to obtain the condition of balance for a Wheatstone Bridge.
- (ii) Use Kirchhoff's rule to determine the currents flowing through the branches MN, TO and SP in the circuit shown in the figure.

5



32. (a) (i) Derive the relation between the current and the drift velocity of free electrons in a conductor. Briefly explain the variation of resistance of a conductor with rise in temperature.
- (ii) An ammeter, together with an unknown resistance in series is connected across two identical batteries, each of emf 1.5 V, connected (i) in series, and (ii) in parallel. If the current recorded in the two cases be $\left(\frac{1}{2}\right)$ A and $\left(\frac{1}{3}\right)$ A respectively, calculate the internal resistance of each battery.

5

OR

1. A heater coil rated as (P, V) is cut into two equal parts. One of the parts is then connected to a battery of V volt. The power consumed by it will be

- (A) P (B) $\frac{P}{2}$
(C) $\frac{P}{4}$ (D) $2P$

28. A potential difference of 1.0 V is applied across a conductor of length 5.0 m and area of cross-section 1.0 mm^2 . When current of 4.25 A is passed through the conductor, calculate
- (i) the drift speed and (ii) relaxation time, of electrons. (Given number density of electrons in the conductor, $n = 8.5 \times 10^{28} \text{ m}^{-3}$).

24. (i) Define 'temperature coefficient of resistance' of a metal.
- (ii) Show the variation of resistivity of copper with rise in temperature.
- (iii) The resistance of a wire is $10\ \Omega$ at $27\ ^\circ\text{C}$. Find its resistance at $-73\ ^\circ\text{C}$.
The temperature coefficient of resistance of the material of the wire is $1.70 \times 10^{-4}\ ^\circ\text{C}^{-1}$.

1. A battery supplies 0.9 A current through a $2\ \Omega$ resistor and 0.3 A current through a $7\ \Omega$ resistor when connected one by one. The internal resistance of the battery is :

(A) $2\ \Omega$

(B) $1.2\ \Omega$

(C) $1\ \Omega$

(D) $0.5\ \Omega$

20. Find the temperature at which the resistance of a conductor increases by 25% of its value at 27 °C. The temperature coefficient of resistance of the conductor is $2.0 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$.

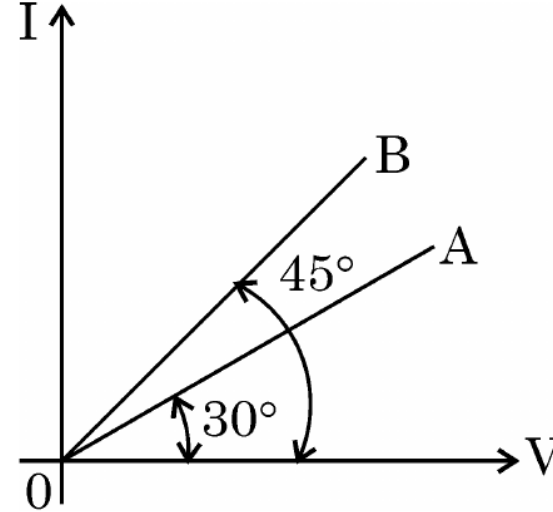
14. **Assertion (A)** : When electrons drift in a conductor, it does not mean that all free electrons in the conductor are moving in the same direction.
Reason (R) : The drift velocity is superposed over large random velocities of electrons.

20. Two wires A and B of different metals have their lengths in ratio 1 : 2 and their radii in ratio 2 : 1 respectively. I-V graphs for them is shown in the figure. Find the ratio of their

2

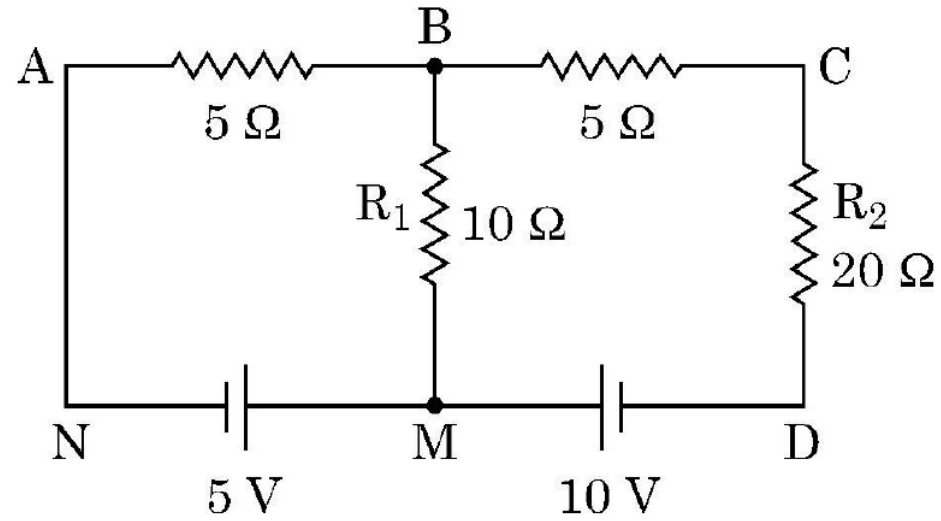
(i) Resistances (R_A/R_B) and

(ii) Resistivities (σ_A/σ_B)



27. Find the currents flowing through the branches AB and BC in the network shown.

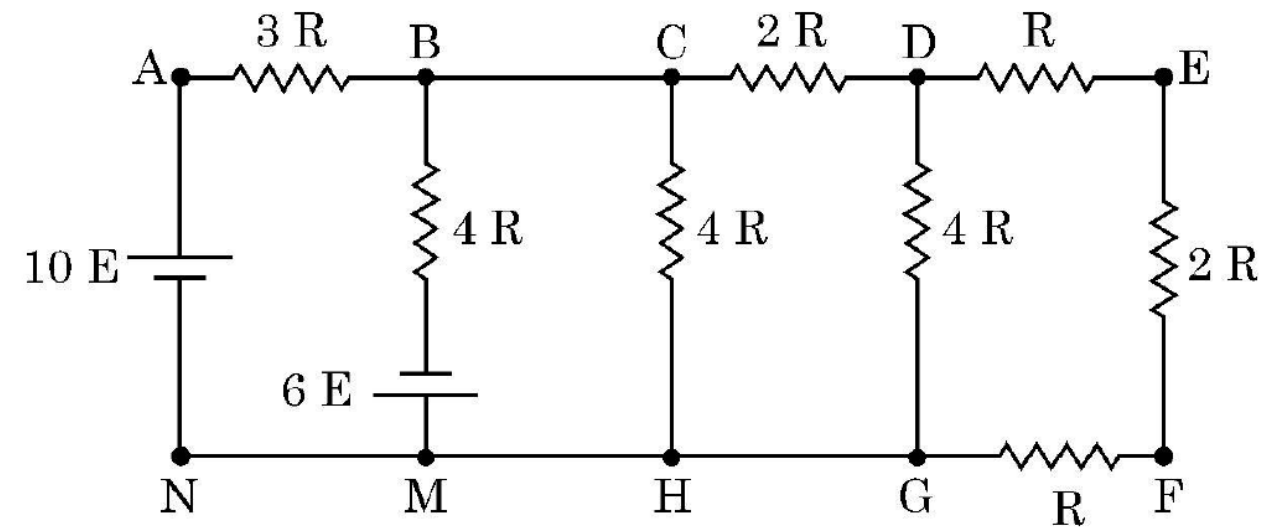
3



20. Two electric heaters have power ratings P_1 and P_2 , at voltage V . They are connected in series to a dc source of voltage V . Find the power consumed by the combination. Will they consume the same power if connected in parallel across the same source ?

25. Find the current in branch BM in the network shown :

3



17. (a) What is meant by 'relaxation time' of free electrons in a conductor ?

Show that the resistance of a conductor can be expressed by

$$R = \frac{m l}{n e^2 \tau A}, \text{ where symbols have their usual meanings.} \quad 2$$

OR

(b) Draw the circuit diagram of a Wheatstone bridge. Obtain the condition when no current flows through the galvanometer in it. 2

22. A current of 1.6 A flows through a wire when a potential difference of 1.0 V is applied across it. The length and cross-sectional area of the wire are 1.0 m and $1.0 \times 10^{-7} \text{ m}^2$ respectively. Calculate :

3

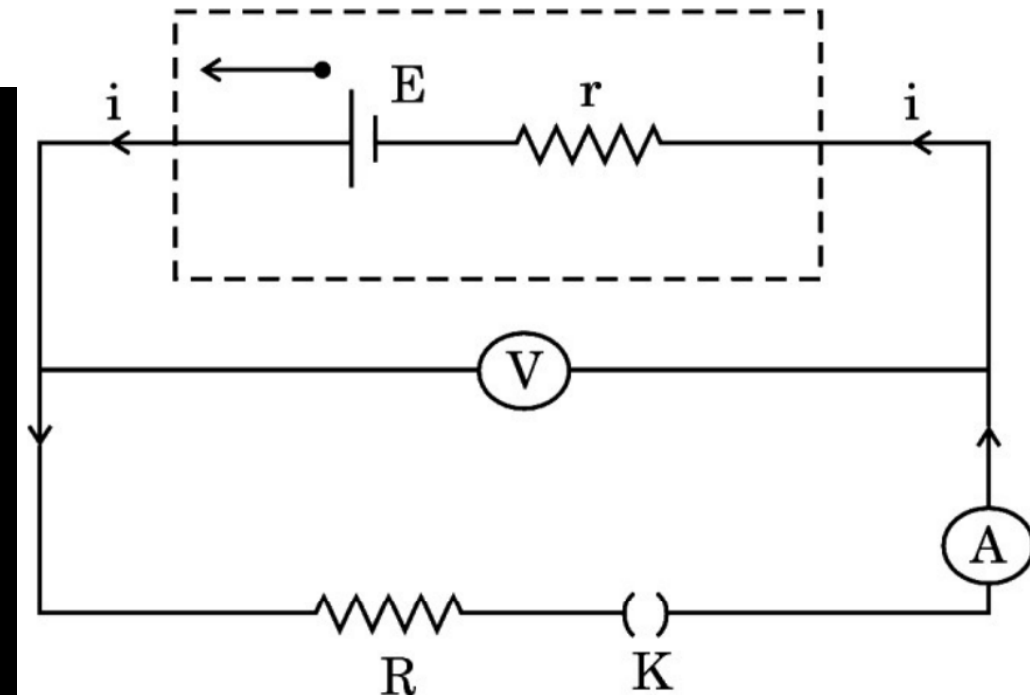
- (a) Electric field across the wire
- (b) Current density
- (c) Average relaxation time (τ)

(Number density of free electrons in the wire is $9.0 \times 10^{28} \text{ m}^{-3}$)

22. A battery of unknown emf E and internal resistance r is connected in a circuit as shown in the figure. When the key (K) is open, the voltmeter reads 10.0 V and ammeter reads 0 A . In the closed circuit, the voltmeter reads 6.0 V and ammeter reads 2.0 A . Calculate :

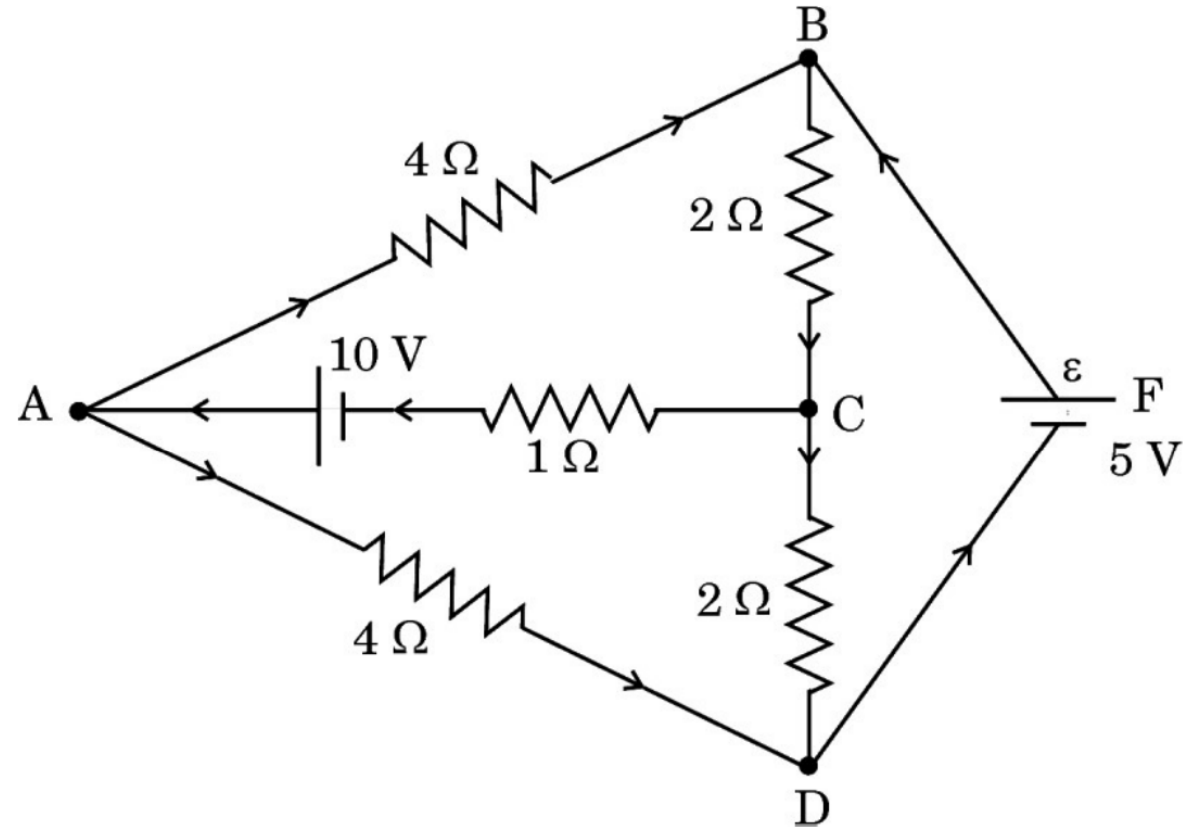
3

- (a) emf of the battery,
- (b) internal resistance of the battery (r), and
- (c) external resistance (R).



22. Determine the current in branches AB, AC and BC of the network shown in figure.

3



2. Electrons drift with speed v_d in a conductor with potential difference V across its ends. If V is reduced to $\left(\frac{V}{2}\right)$, their drift speed will become :

(A) $\frac{v_d}{2}$

(B) v_d

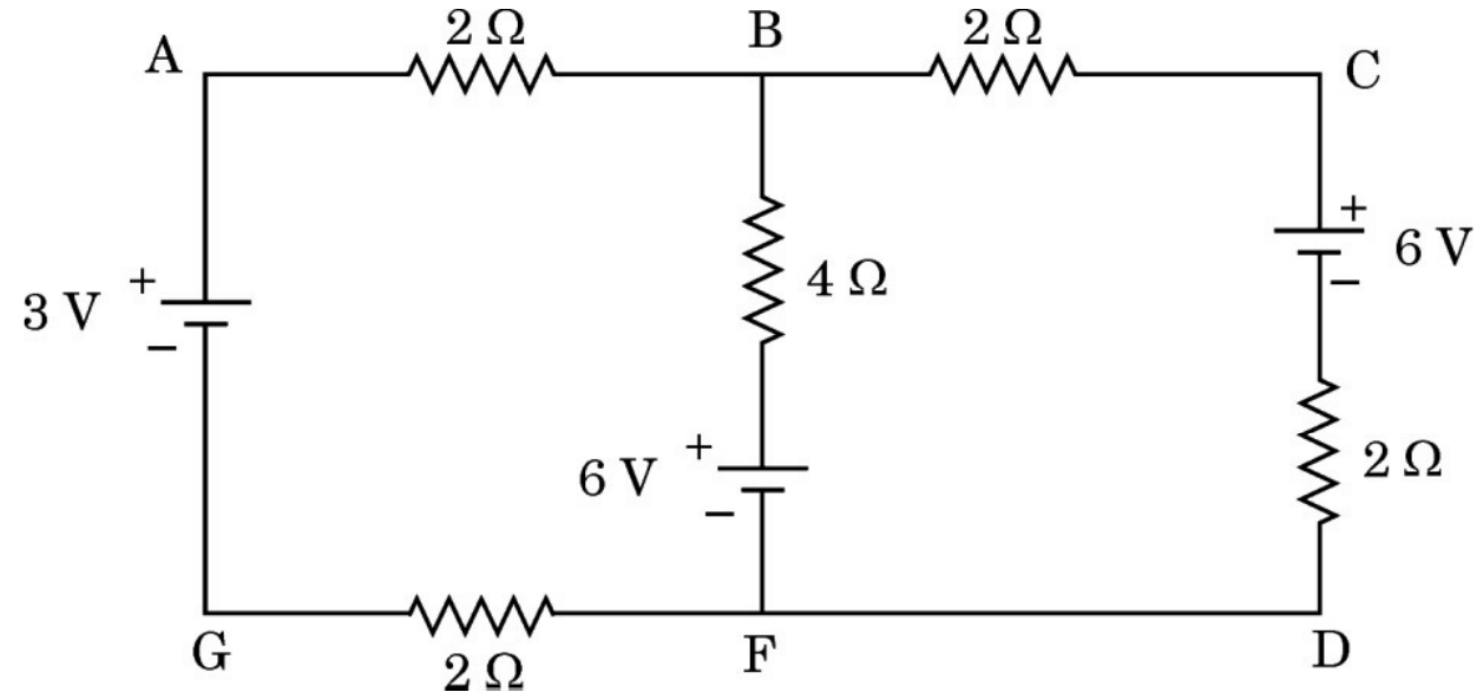
(C) $2 v_d$

(D) $4 v_d$

17. An electric field E is maintained in a wire of length ' l ' and area of cross-section ' a '. Derive the relation between the current density ' σ ' in the wire and the electric field E .

25. The figure shows a circuit with three ideal batteries. Find the magnitude and direction of currents in the branches AG, BF and CD.

3



30. When the terminals of a cell are connected to a conductor of resistance R , an electric current flows through the circuit. The electrolyte of the cell also offers some resistance in the path of the current, like the conductor. This resistance offered by the electrolyte is called internal resistance of the cell (r). It depends upon the nature of the electrolyte, the area of the electrodes immersed in the electrolyte and the temperature. Due to internal resistance, a part of the energy supplied by the cell is wasted in the form of heat.

When no current is drawn from the cell, the potential difference between the two electrodes is known as emf of the cell (ϵ). With a current drawn from the cell, the potential difference between the two electrodes is termed as terminal potential difference (V).

- (i) Choose the ***incorrect*** statement : 1
- (A) The potential difference (V) between the two terminals of a cell in a closed circuit is always less than its emf (ϵ), during discharge of the cell.
 - (B) The internal resistance of a cell decreases with the decrease in temperature of the electrolyte.
 - (C) When current is drawn from the cell then $V = \epsilon - Ir$.
 - (D) The graph between potential difference between the two terminals of the cell (V) and the current (I) through it is a straight line with a negative slope.
- (ii) Two cells of emfs 2.0 V and 6.0 V and internal resistances 0.1 Ω and 0.4 Ω respectively, are connected in parallel. The equivalent emf of the combination will be : 1
- (A) 2.0 V
 - (B) 2.8 V
 - (C) 6.0 V
 - (D) 8.0 V

(iii) Dipped in the solution, the electrode exchanges charges with the electrolyte. The positive electrode develops a potential V_+ ($V_+ > 0$), and the negative electrode develops a potential $- (V_-)$ ($V_- \geq 0$), relative to the electrolyte adjacent to it. When no current is drawn from the cell then :

- | | |
|-----------------------------------|-----------------------------------|
| (A) $\varepsilon = V_+ + V_- > 0$ | (B) $\varepsilon = V_+ - V_- > 0$ |
| (C) $\varepsilon = V_+ + V_- < 0$ | (D) $\varepsilon = V_+ + V_- = 0$ |

(iv) (a) Five identical cells, each of emf 2 V and internal resistance 0.1Ω are connected in parallel. This combination in turn is connected to an external resistor of 9.98Ω . The current flowing through the resistor is :

- | | |
|------------|-----------|
| (A) 0.05 A | (B) 0.1 A |
| (C) 0.15 A | (D) 0.2 A |

OR

(b) Potential difference across a cell in the open circuit is 6 V. It becomes 4 V when a current of 2 A is drawn from it. The internal resistance of the cell is :

- | | |
|------------------|------------------|
| (A) 1.0Ω | (B) 1.5Ω |
| (C) 2.0Ω | (D) 2.5Ω |

17. (a) “The electron drift speed is only a few mm/s for currents in the range of a few amperes for a given conductor.” How then is current established almost the instant a circuit is closed ? Explain.
- (b) ‘ $V = IR$ is a statement of Ohm’s Law’ is not true. Explain.

17. Define resistivity of a conductor. How does the resistivity of a conductor depend upon the following :

2

- (a) Number density of free electrons in the conductor (n)
- (b) Their relaxation time (τ)

3. In a uniform straight wire, conduction electrons move along + x direction. Let $\vec{E}$ and $\vec{j}$ be the electric field and current density in the wire, respectively. Then :
- (A) $\vec{E}$ and $\vec{j}$ both are along + x direction.
 - (B) $\vec{E}$ and $\vec{j}$ both are along - x direction.
 - (C) $\vec{E}$ is along + x direction, but $\vec{j}$ is along - x direction.
 - (D) $\vec{E}$ is along - x direction, but $\vec{j}$ is along + x direction.

17. A uniform wire of length L and area of cross-section A has resistance R . The wire is uniformly stretched so that its length increases by 25%. Calculate the percentage increase in the resistance of the wire.

- 23.** (a) Define current density. Is it a scalar or a vector ? An electric field $\vec{E}$ is maintained in a metallic conductor. If n be the number of electrons (mass m , charge $-e$) per unit volume in the conductor and τ its relaxation time, show that the current density $\vec{j} = \alpha \vec{E}$, where $\alpha = \left(\frac{ne^2}{m} \right) \tau$. 3

OR

- (b) What is a Wheatstone bridge ? Obtain the necessary conditions under which the Wheatstone bridge is balanced. 3

3. A student is asked to connect four cells, each of emf E and internal resistance r , in series. But she/he connects one cell wrongly in series with the other cells. The equivalent emf and the equivalent internal resistance of the combination will be :

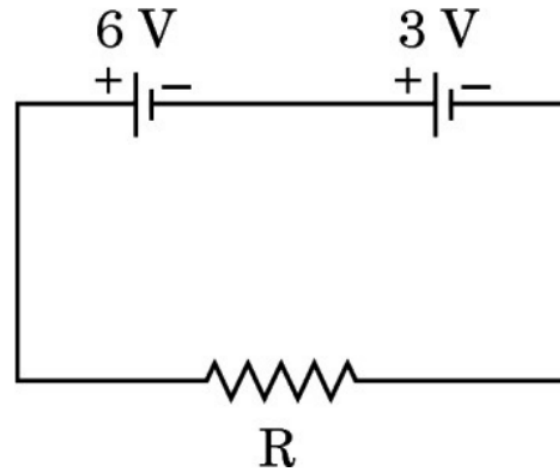
(A) $4E$ and $2r$

(B) $4E$ and $3r$

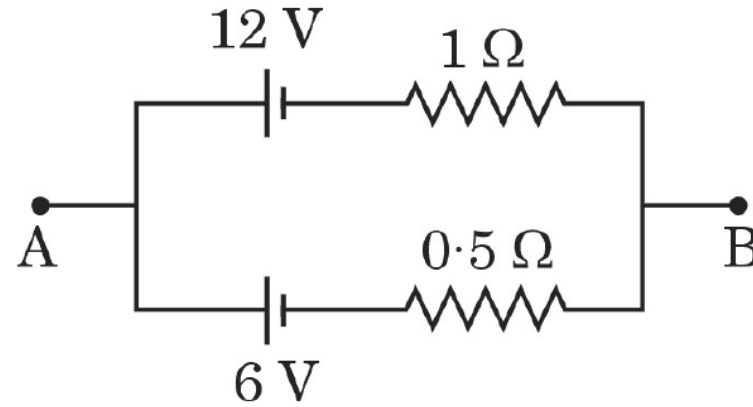
(C) $3E$ and $4r$

(D) $2E$ and $4r$

17. Two batteries of emfs 6 V and 3 V and internal resistances $0.8\ \Omega$ and $0.2\ \Omega$ respectively are connected in series to an external resistance R , as shown in figure. Find the value of R so that the potential difference across the 6 V battery be zero.



3. Consider the circuit shown in the figure. The potential difference between points A and B is :



(A) 6 V

(B) 8 V

(C) 9 V

(D) 12 V

17. Find the temperature at which the resistance of a wire made of silver will be twice its resistance at 20°C . Take 20°C as the reference temperature and temperature coefficient of resistance of silver at $20^{\circ}\text{C} = 4.0 \times 10^{-3} \text{ K}^{-1}$.

2. The emf and internal resistance of a cell are 3 V and $0.2\ \Omega$ respectively. It is connected to an external resistor of $5.8\ \Omega$. The potential difference across the cell will be :
- (A) 1.2 V (B) 2.0 V
(C) 2.9 V (D) 3.2 V

17. Two identical cells, each of emf E and internal resistance r , are connected in series with a resistor of $8\ \Omega$. A current of 1.5 A flows in the circuit. When these cells are connected in parallel, they send 1.0 A current through the same resistor. Calculate the internal resistance, r and emf, E of the cell.

- 22.** (a) Define the terms – ‘temperature coefficient of resistance’ and ‘electrical conductivity’ of a conductor. Why are constantan and manganin used for making standard resistances ? Explain. 3

OR

- (b) Define ‘drift velocity’. Show that resistivity of the material of a conductor is inversely proportional to the relaxation time for the free electrons in the conductor. 3

19. (a) A cell is connected across an external resistance $12\ \Omega$ and supplies $0.25\ \text{A}$ current. When the external resistance is increased by $4\ \Omega$, the current reduces to $0.2\ \text{A}$. Calculate (i) the emf, and (ii) the internal resistance, of the cell.

15. *Assertion (A) :* The current density ($\vec{J}$) at a point in a conducting wire is in the direction of electric field ($\vec{E}$) at that point.

Reason (R) : A conducting wire obeys Ohm's law.

28. (a) The radius of a conducting wire AB uniformly decreases from its one end A to another end B. It is connected across a battery. How will (i) electric field, (ii) current density, and (iii) mobility of electrons change from end A to end B ? Justify your answer in each case.

22. (a) Define mobility of free electrons in a conductor. How does the mobility of electrons in a metal change when :

(i) temperature of the conductor is decreased at constant potential difference ?

(ii) applied potential difference is doubled at a constant temperature ?

3

OR

(b) Obtain an expression for emf and internal resistance of a cell which is equivalent to parallel combination of two cells of emf and internal resistance (E_1, r_1) and (E_2, r_2), respectively.

3

17. Two conducting wires of same material have radii r and $\frac{r}{2}$. The current flowing through them is I and $2I$, respectively. Find the ratio of drift velocity of free electrons in the wires.

3. A steady current I is passed through a conductor at room temperature for time t . It is observed that its temperature rises by 0.5°C . If $2I$ current is passed through the conductor (at room temperature) for the same duration, the rise in its temperature will be approximately :

(A) 1.0°C

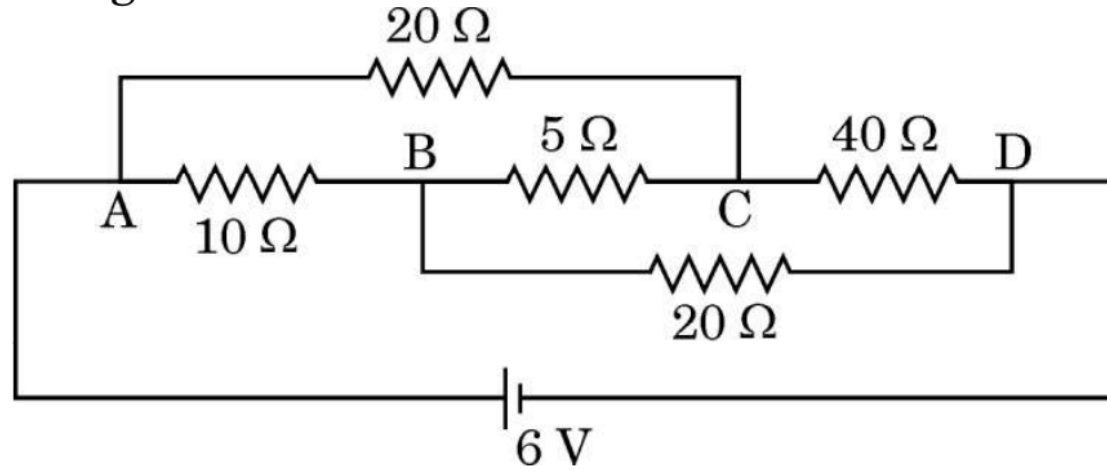
(B) 1.5°C

(C) 2.0°C

(D) 4.0°C

17. Calculate the value of the current passing through the battery in the given circuit diagram.

2



29. In a metallic conductor, an electron, moving due to thermal motion, suffers collisions with the heavy fixed ions but after collision, it will emerge out with the same speed but in random directions. If we consider all the electrons, their average velocity will be zero. When an electric field is applied, electrons move with an average velocity, known as drift velocity (v_d). The average time between successive collisions is known as relaxation time (τ). The magnitude of drift velocity per unit electric field is called mobility (μ).

An expression for current through the conductor can be obtained in terms of drift velocity, number of electrons per unit volume (n), electronic charge ($-e$), and the cross-sectional area (A) of the conductor. This expression leads to an expression between current density ($\hat{j}$) and the electric field ($\vec{E}$). Hence, an expression for resistivity (ρ) of a metal is obtained. This expression helps us to understand increase in resistivity of a metal with increase in its temperature, in terms of change in the relaxation time (τ) and change in the number density of electrons (n).

- (i) (a) Consider the contribution of the following two factors I and II in resistivity of a metal :

I. Relaxation time of electrons

II. Number of electrons per unit volume

The resistivity of a metal increases with increase in its temperature because :

- (A) I decreases and II increases.
- (B) I increases and II is almost constant.
- (C) Both I and II increase.
- (D) I decreases and II is almost constant.

OR

(b) A steady current flows in a copper wire of non-uniform cross-section. Consider the following three physical quantities :

- I. Electric field
- II. Current density
- III. Drift speed

Then at the different points along the wire :

1

- (A) II and III change, but I is constant.
- (B) I and II change, but III is constant.
- (C) I and III change, but II is constant.
- (D) All I, II and III change.

(ii) The temperature coefficient of resistance of nichrome is $1.70 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$. In order to increase resistance of a nichrome wire by 8.5%, the temperature of the wire should be increased by : 1

(A) 250°C

(B) 500°C

(C) 850°C

(D) 1000°C

(iii) A wire of length 0.5 m and cross-sectional area $1.0 \times 10^{-7} \text{ m}^2$ is connected to a battery of 2 V that maintains a current of 1.5 A in it. The conductivity of the material of the wire (in $\Omega^{-1} \text{ m}^{-1}$) is : 1

(A) 2.5×10^4

(B) 3.0×10^5

(C) 3.75×10^6

(D) 5.0×10^7

(iv) Consider two cylindrical conductors A and B, made of the same metal connected in series to a battery. The length and the radius of B are twice that of A. If μ_A and μ_B are the mobility of electrons in A and B respectively, then $\frac{\mu_A}{\mu_B}$ is :

1

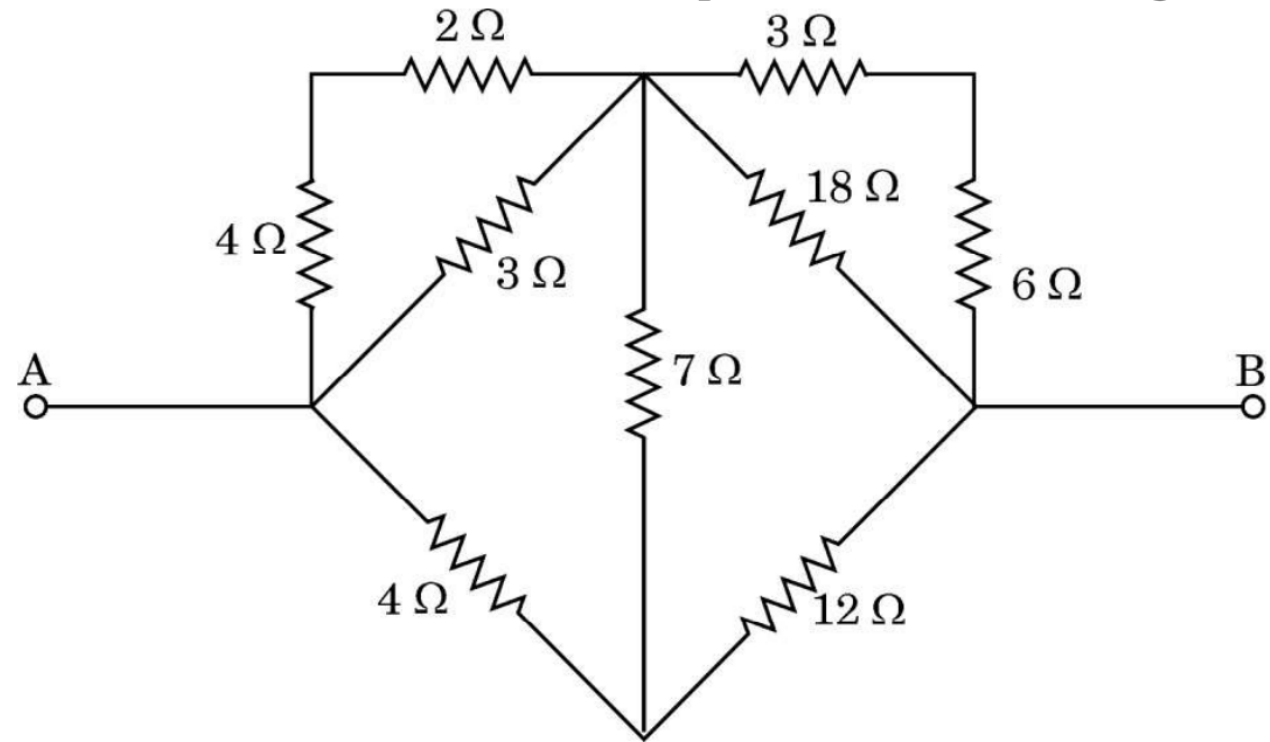
(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) 2

(D) 1

7. The effective resistance between points A and B in the given circuit is :



(A) $6\ \Omega$

(B) $\frac{8}{3}\ \Omega$

(C) $\frac{16}{3}\ \Omega$

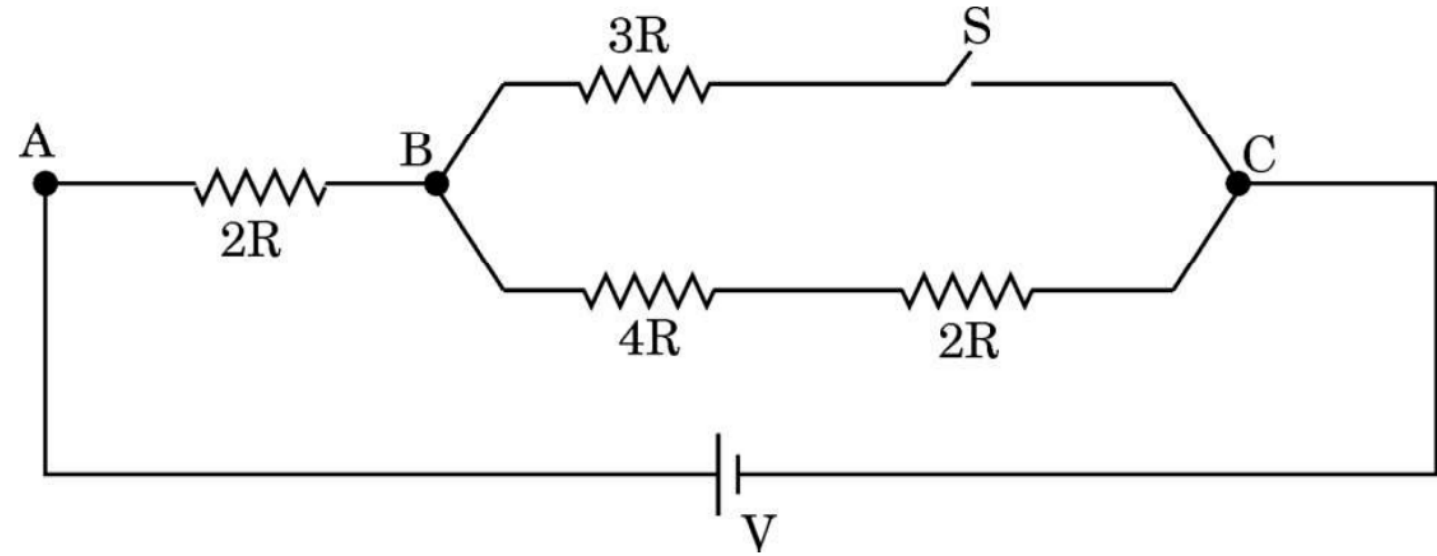
(D) $2\ \Omega$

17. Two wires made of the same material have the same length (l) but different cross-sectional areas A_1 and A_2 . They are connected together with a cell of voltage V . Find the ratio of the drift velocities of free electrons in the two wires when they are joined in (i) series, and (ii) parallel.

- 17.** A cell of emf E and internal resistance r is connected across a resistor of variable resistance R . Show graphically the variation of
- (a) the terminal voltage across the cell,
 - (b) the current supplied by the cell,
- with R as it is increased from 0 to the maximum value.

- 22.** Three cells A, B and C of emfs 2 V, 3 V and 5 V respectively are connected in parallel to each other. Their internal resistances are $5\ \Omega$, $5\ \Omega$ and $1\ \Omega$ respectively. Calculate the currents flowing through the cells A, B and C. 3

1. The ratio of potential difference across AB in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is :



(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

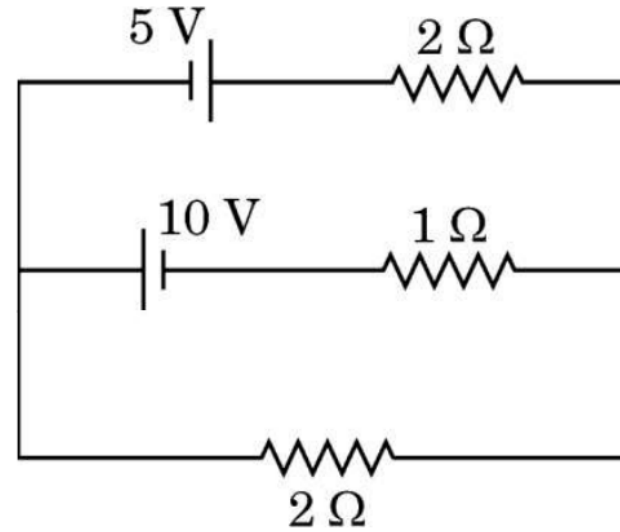
(C) 1

(D) 2

18. (a) What is the difference between 'velocity' and 'drift velocity' of electrons in a current-carrying conductor.
- (b) A copper wire of uniform cross-sectional area carries a current of 3.4 A. The drift velocity of conduction electrons is 0.2 mm/s. If the number density of electrons in copper is $8.5 \times 10^{28} \text{ m}^{-3}$, find the area of cross-section of the wire.

- 26.** State Kirchhoff's laws. Apply these laws to find the values of current flowing in the three branches of the given circuit.

3



1. A wire of resistance R , connected to an ideal battery consumes a power P . If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be :

(A) $\frac{P}{4}$

(B) $\frac{P}{2}$

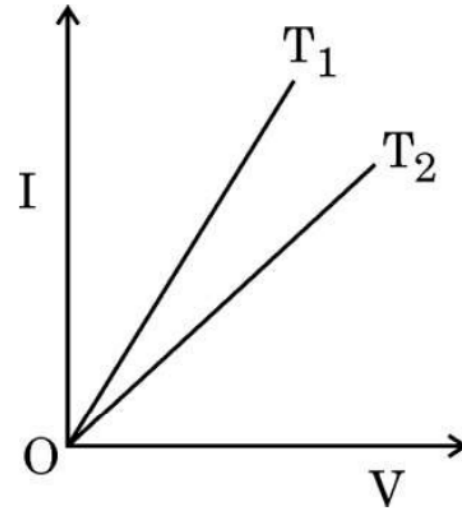
(C) P

(D) $2P$

17. A cell of emf E and internal resistance r is connected to an external variable resistance R . Plot a graph showing the variation of terminal voltage V of the cell as a function of current I , supplied by the cell. Explain how the emf of the cell and its internal resistance can be found from it.

- 22.** (a) Define the term 'drift velocity' of conduction electrons in a conductor.
- (b) A conductor of length l and area of cross-section A is connected across an ideal battery of emf V . Derive the formula for the current density in terms of relaxation time τ .

1. The figure shows the voltage (V) versus the current (I) graphs for a wire at two temperatures T_1 and T_2 . One can conclude that :



(A) $T_2 = 2T_1$

(B) $T_1 > T_2$

(C) $T_1 = T_2 / 3$

(D) $T_1 < T_2$

2. If R_s and R_p are the equivalent resistances of n resistors, each of value R , in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is :

(A) $\left(\frac{n^2 - 1}{n^2}\right)R$

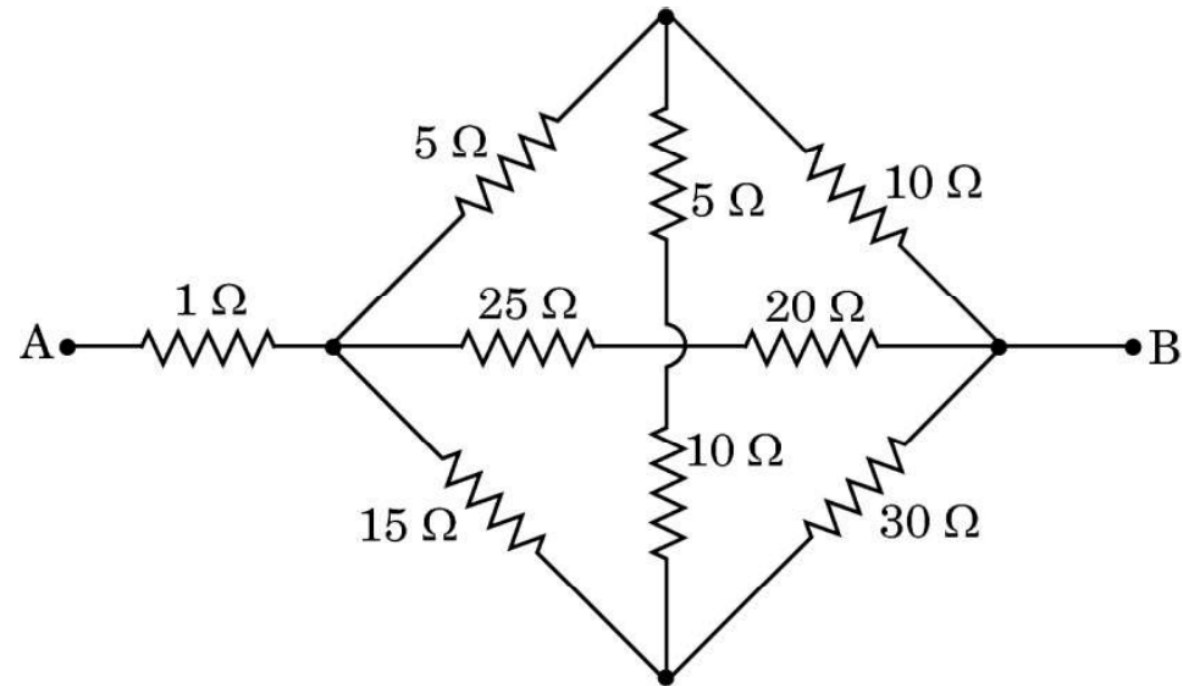
(B) $\left(\frac{n^2 + 1}{n^2 - 1}\right)R$

(C) $\left(\frac{n^2 - 1}{n}\right)R$

(D) $\frac{(n^2 + 1)R}{n^2}$

17. Find the equivalent resistance between points A and B for the network shown in the figure.

2



2. The resistance of a wire at $20\text{ }^{\circ}\text{C}$ is $50\ \Omega$. When it is heated to $120\text{ }^{\circ}\text{C}$, the resistance becomes $51\ \Omega$. The temperature coefficient of resistance of the material of the wire is **1**

(A) $2 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$

(B) $1 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$

(C) $5 \times 10^{-4}\text{ }^{\circ}\text{C}^{-1}$

(D) $1 \times 10^{-3}\text{ }^{\circ}\text{C}^{-1}$

17. A 20.0 cm long wire is connected across an ideal battery of 4 V. The drift velocity of conduction electrons in the wire is 0.8 mm/s. Calculate the relaxation time of the electrons.

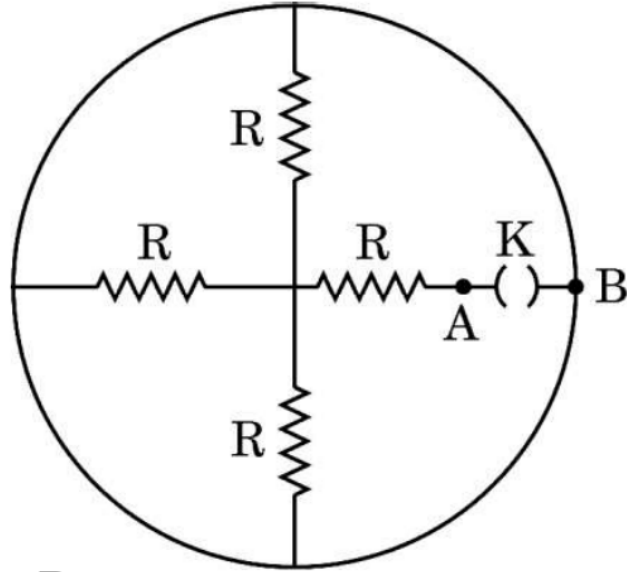
2

22. (a) Define the terms 'emf' and 'terminal voltage' of a cell. Can the terminal voltage of a cell ever exceed its emf ? Explain. **3**

OR

- (b) Define the term current density. Write its SI unit. Derive the equivalent form of Ohm's law $\vec{j} = \sigma \vec{E}$. **3**

5. Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open, will be :



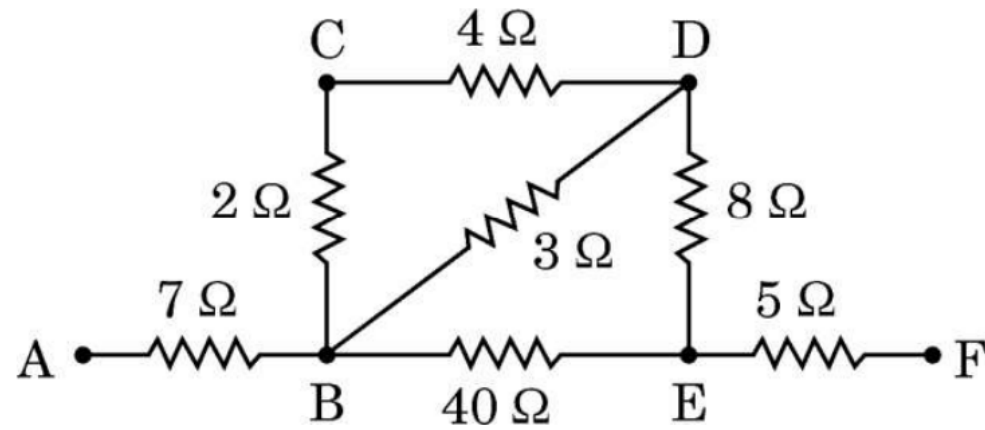
(A) $4R$

(B) ∞

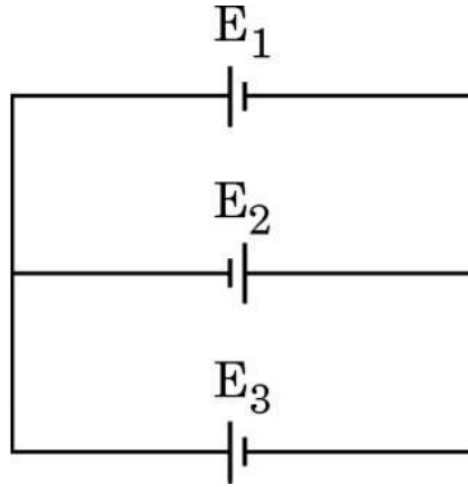
(C) $\frac{R}{4}$

(D) $\frac{4R}{3}$

17. Find the effective resistance of the network of resistors between points A and F as shown in the figure.

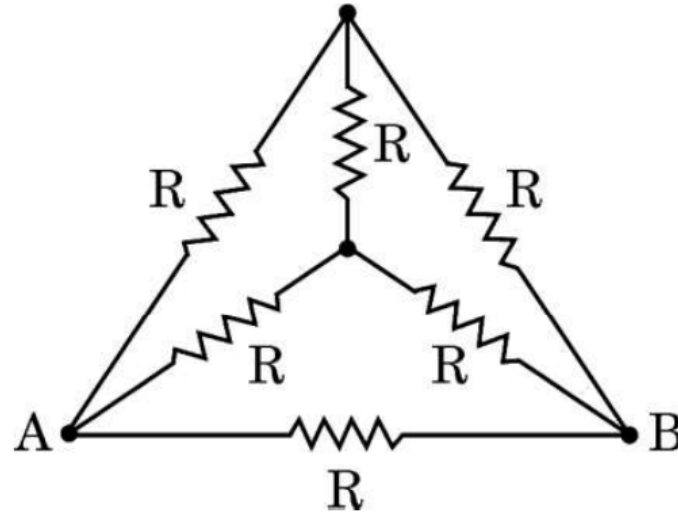


- 32.** (a) (i) Three batteries E_1 , E_2 and E_3 of emfs and internal resistances (4 V, $2\ \Omega$), (2 V, $4\ \Omega$) and (6 V, $2\ \Omega$) respectively are connected as shown in the figure. Find the values of the currents passing through batteries E_1 , E_2 and E_3 .

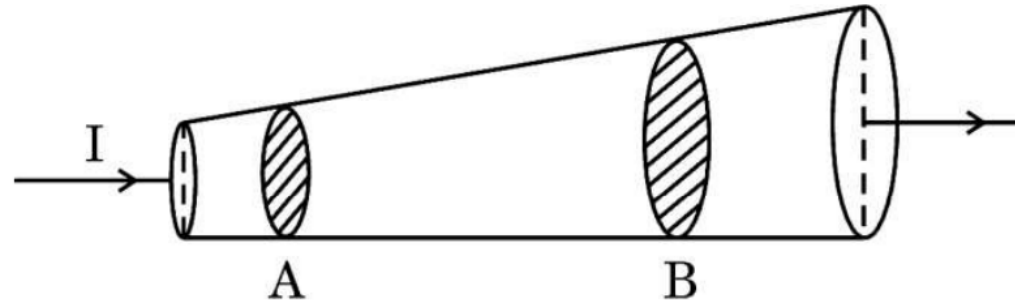


- (ii) The ends of six wires, each of resistance R ($= 10\ \Omega$) are joined as shown in the figure. The points A and B of the arrangement are connected in a circuit. Find the value of the effective resistance offered by it to the circuit.

5



- (b) (i) Current I ($= 1 \text{ A}$) is passing through a copper rod ($n = 8.5 \times 10^{28} \text{ m}^{-3}$) of varying cross-sections as shown in the figure. The areas of cross-section at points A and B along its length are $1.0 \times 10^{-7} \text{ m}^2$ and $2.0 \times 10^{-7} \text{ m}^2$ respectively. Calculate :



- (I) the ratio of electric fields at points A and B.
- (II) the drift velocity of free electrons at point B.

- 18.** A wire of resistance X ohm is gradually stretched till its length becomes twice its original length. If its new resistance becomes $40\ \Omega$, find the value of X .

- 21.** The resistance of a wire at 25°C is $10.0\ \Omega$. When heated to 125°C , its resistance becomes $10.5\ \Omega$. Find (i) the temperature coefficient of resistance of the wire, and (ii) the resistance of the wire at 425°C . 2

2. Three wires A, B and C of the same material have lengths and area of cross-sections as $(2l, \frac{A}{2})$, (l, A) and $(\frac{l}{2}, 2A)$, respectively. If the resistances of these wires are R_A , R_B and R_C respectively, then :

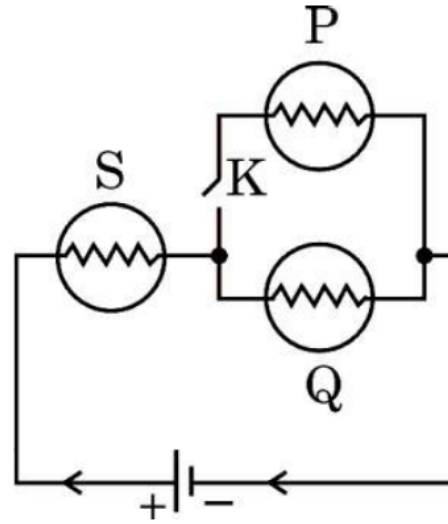
(A) $R_A > R_B > R_C$

(B) $R_B > R_C > R_A$

(C) $R_C > R_A > R_B$

(D) $R_A > R_C > R_B$

21. (a) In the given figure, three identical bulbs P, Q and S are connected to a battery.

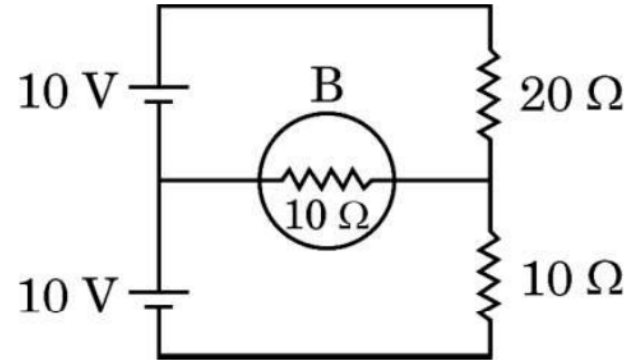


- (i) Compare the brightness of bulbs P and Q with that of bulb S when key K is closed.
- (ii) Compare the brightness of the bulbs S and Q when the key K is opened.

Justify your answer in both cases.

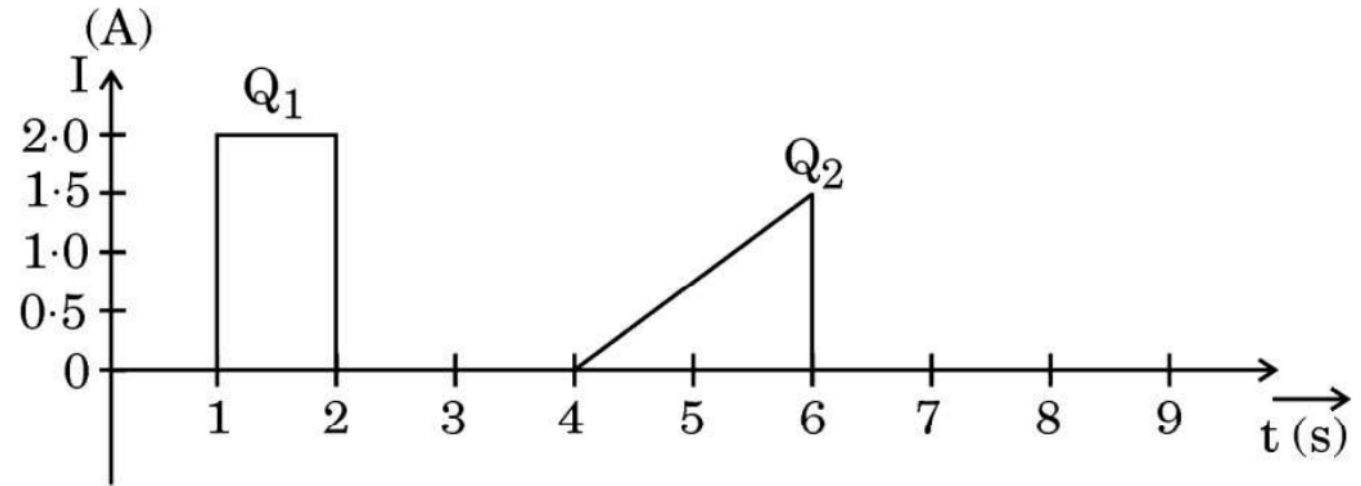
OR

- (b) Two cells of emf 10 V each, two resistors of $20\ \Omega$ and $10\ \Omega$ and a bulb B of $10\ \Omega$ resistance are connected together as shown in the figure. Find the current that flows through the bulb.



27. (a) (i) Derive an expression for the resistivity of a conductor in terms of number density of free electrons and relaxation time.
- (ii) The figure shows the plot of current through a cross-section of wire over two different time intervals. Compare the charges (Q_1 and Q_2) that pass through the cross-section during these time intervals.

3



OR

- (b) (i) A battery of emf E and internal resistance r is connected to a variable external resistance R .
- (I) Obtain the expression for current I in the circuit and the value of maximum current the battery can supply.
- (II) Obtain the terminal voltage V across the battery and its maximum possible value.
- (ii) The above battery sends a current I_1 when $R = R_1$ and a current I_2 when $R = R_2$. Obtain the internal resistance of the battery in terms of I_1 , I_2 , R_1 and R_2 .

2. When the switch of the circuit is turned on, the filament of the bulb glows instantaneously because :
- (A) the electrons coming from the power source move fast through the initially empty filament.
 - (B) the filament may be old having low resistance.
 - (C) electric field is established instantaneously across the filament which pushes the electrons.
 - (D) free electrons in the filament travel with the speed of light.

2. A current flows through a cylindrical conductor of radius R . The current density at a point in the conductor is $j = \alpha r$ (along its axis), here α is a constant and r is distance from the axis of the conductor. The current flowing through the portion of the conductor from $r = 0$ to $r = \frac{R}{2}$ is proportional to :

(A) R

(B) R^2

(C) R^3

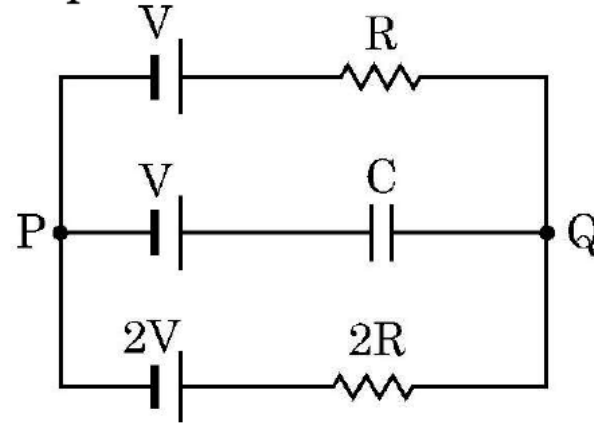
(D) R^4

21. (a) Two wires of the same material and the same radius have their lengths in the ratio $2 : 3$. They are connected in parallel to a battery which supplies a current of 15 A . Find the current through the wires.

2

OR

- (b) In the circuit three ideal cells of e.m.f. V , V and $2V$ are connected to a resistor of resistance R , a capacitor of capacitance C and another resistor of resistance $2R$ as shown in figure. In the steady state find (i) the potential difference between P and Q and (ii) potential difference across capacitor C .



22. (a) Define Electrical conductivity. Obtain the expression of electrical conductivity of a conductor in terms of number density and relaxation time of free electrons.
- (b) Explain qualitative change in resistivity of a conductor with temperature using expression obtained in (a).

2. A battery of e.m.f. 12 V and internal resistance $0.5\ \Omega$ is connected to a $9.5\ \Omega$ resistor through a key. The ratio of potential difference between the two terminals of the battery, when the key is open to that when the key is closed, is

(A) 1.05

(B) 1

(C) 0.95

(D) 1.1

22. (a) Define resistivity of a conductor. Discuss its dependence on temperature of the conductor and draw a plot of resistivity of copper as a function of temperature.

3

- (b) (i) “A low voltage battery from which high current is required must have low internal resistance.” Justify.
- (ii) “A high voltage battery must have a large internal resistance.” Justify.

2. A student has three resistors, each of resistance R . To obtain a resistance of $\frac{2}{3} R$, she should connect

1

- (A) all the three resistors in series.
- (B) all the three resistors in parallel.
- (C) two resistors in series and then this combination in parallel with the third resistor.
- (D) two resistors in parallel and then this combination in series with the third resistor.

22. (a) A cell of e.m.f. E and internal resistance r is connected with a variable external resistance R and a voltmeter showing potential drop V across R . Obtain the relationship between V , E , R and r .
- (b) Draw the shape of the graph showing the variation of terminal voltage V of the cell as a function of current I drawn from it. How one can determine the e.m.f. of the cell and its internal resistance from this graph ?

2. Two conductors A and B of the same material have their lengths in the ratio 1:2 and radii in the ratio 2:3. If they are connected in parallel across a battery, the ratio $\frac{v_A}{v_B}$ of the drift velocities of electrons in them will be - 1

- | | |
|-------------------|-------------------|
| (A) 2 | (B) $\frac{1}{2}$ |
| (C) $\frac{3}{2}$ | (D) $\frac{8}{9}$ |

17. n identical cells, each of e.m.f. E and internal resistance r , are connected in series. Later on it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'. 2

22. (a) Two batteries of emf's 3V & 6V and internal resistances $0.2\ \Omega$ & $0.4\ \Omega$ are connected in parallel. This combination is connected to a $4\ \Omega$ resistor. Find :

3

- (i) the equivalent emf of the combination
- (ii) the equivalent internal resistance of the combination
- (iii) the current drawn from the combination

OR

- (b) (i) A conductor of length l is connected across an ideal cell of emf E . Keeping the cell connected, the length of the conductor is increased to $2l$ by gradually stretching it. If R and R' are initial and final values of resistance and v_d and v_d' are initial and final values of drift velocity, find the relation between (i) R' and R and (ii) v_d' and v_d .
- (ii) When electrons drift in a conductor from lower to higher potential, does it mean that all the 'free electrons' of the conductor are moving in the same direction ?

2. The resistance of a wire of length L and radius r is R . Which one of the following would provide a wire of the same material of resistance $\frac{R}{2}$? 1

- (A) Using a wire of same radius and twice the length
- (B) Using a wire of same radius and half length
- (C) Using a wire of same length and twice the radius
- (D) Using a wire of same length and half the radius

17. Show that $\vec{E} = -\rho \vec{j}$ leads to Ohm's law. Write a condition in which the Ohm's law is not valid for a material.

17. A battery of emf E and internal resistance r is connected to a rheostat. When a current of $2A$ is drawn from the battery, the potential difference across the rheostat is $5V$. The potential difference becomes $4V$ when a current of $4A$ is drawn from the battery. Calculate the value of E and r .